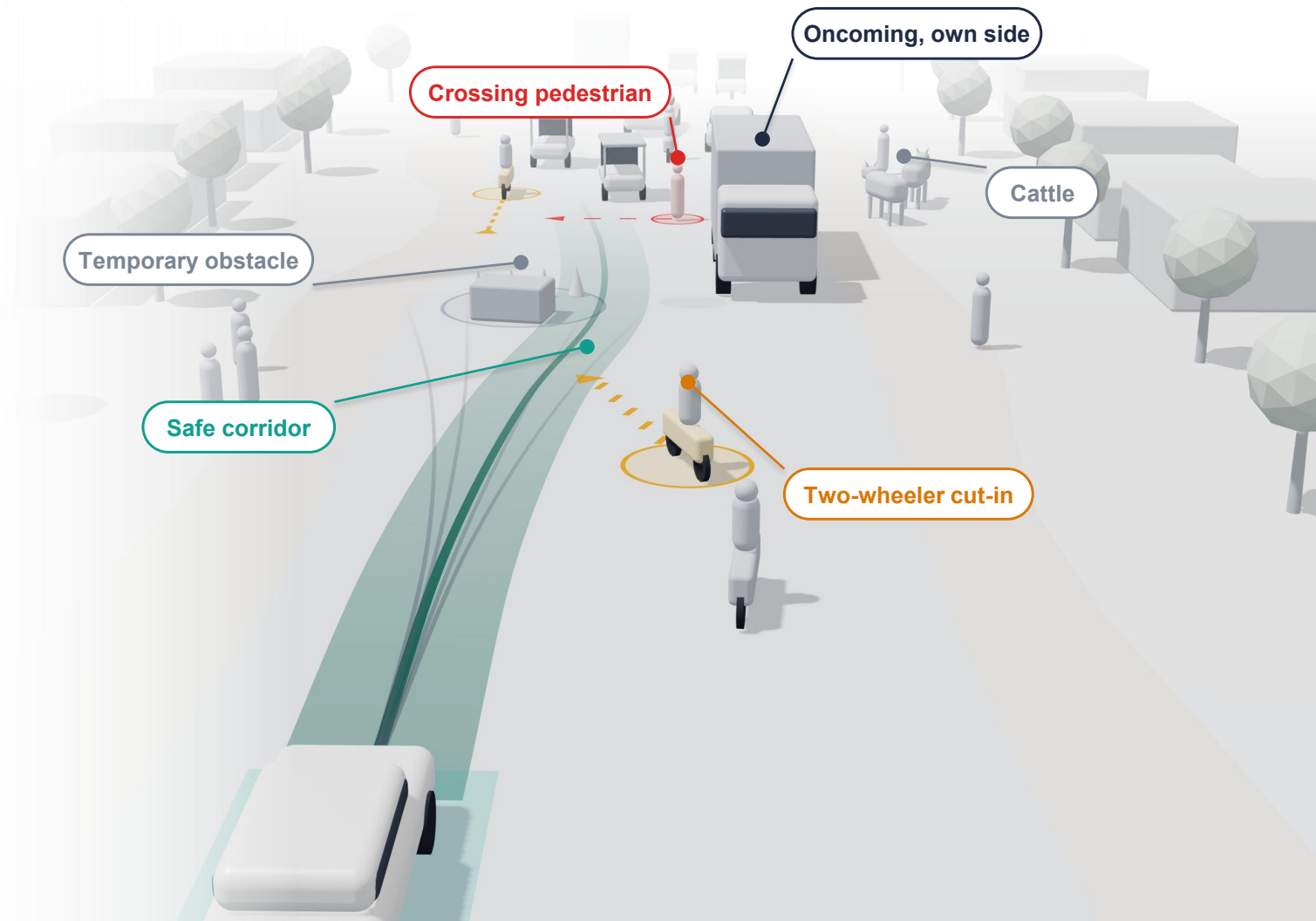


TITLE PAGE

PATHSENSE

Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

- **Problem Statement ID** – SIH26037
- **Problem Statement Title**- Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme**- Smart Vehicles
- **PS Category**- Software
- **Team ID**-
- **Team Name** – Ctrl Alt Elite



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

See every road user, predict where they are going, and keep re-planning a collision-free path — without relying on lane markings.

Detailed explanation of the proposed solution

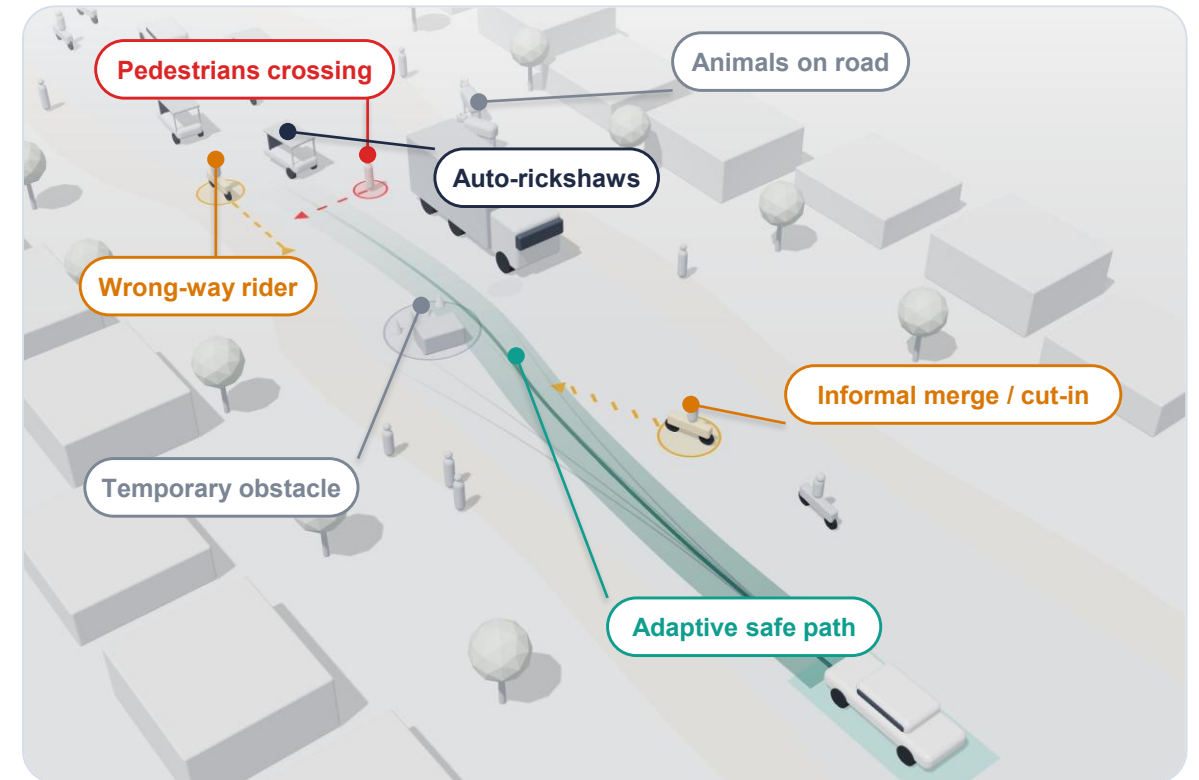
- Camera-first perception detects and tracks cars, buses, trucks, auto-rickshaws, two-wheelers, pedestrians and animals
- Short-term motion prediction + a bird's-eye risk map of the scene
- Adaptive planner samples candidate paths and keeps the safest collision-free one, every frame

How it addresses the problem

- Plans in drivable free space, so missing lane markings are not a blocker
- Reads cut-ins, crossings, wrong-way and sudden turns from motion history
- Built for left-hand, mixed Indian traffic

Innovation and uniqueness of the solution

- Threat to the ego path, not mere proximity: traffic on its own side and people on the footpath are watched; anything entering the path is acted on
- Explainable: every GO / SLOW / BRAKE carries its reason



Also designed for:

Mixed traffic

No lane markings

Two-wheelers

Sudden turns

Technologies to be used (e.g. programming languages, frameworks, hardware)

Perception	Prediction & planning	Software	Sensors & compute	Simulation & validation · pilot stage
Object detection (YOLO family)	Short-term motion prediction	Python · PyTorch	Front camera (prototype)	MATLAB / Simulink
Multi-object tracking	Bird's-eye risk / cost map	OpenCV	Radar / LiDAR (pilot)	RoadRunner scenarios
Monocular depth estimation	Kinematic path sampling	Web dashboard for review	Automotive compute (production)	Automated Driving Toolbox

Methodology and process for implementation (Flow Charts/Images/ working prototype)**Lane-free planning**

Plans in drivable free space, so missing or faded lane markings don't break it.

**Threat, not proximity**

Traffic on its own side and footpath pedestrians are watched; agents entering the path trigger action.

**Modular by design**

Perception, prediction and planning are separate stages — sensors and simulators plug in later.

Analysis of the feasibility of the idea

✓ FEASIBILITY

- Builds on established perception, tracking and motion-planning methods
- Camera-first design gives an accessible first deployment
- Modular: LiDAR / radar can be added in later stages
- Validation grows from recorded roads to simulation and closed-loop tests
- Mature automotive simulation and perception ecosystems support scale-up

🚀 VIABILITY

- Designed around real Indian-road traffic complexity
- Fits urban roads, village roads, highways and dense mixed traffic
- Modular perception / planning supports progressive upgrades
- Evolves from research prototype → pilot → production
- Can integrate with vehicle systems and mobility platforms

Potential challenges and risks

⚠️ **Unstructured roads, missing lane markings**



Ego-path estimation and scene-aware planning instead of lane-only planning

⚠️ **Irregular traffic behaviour**



Short-term motion prediction and dynamic risk assessment

⚠️ **Perception uncertainty**



Confidence-aware decisions, temporal tracking, future multi-sensor fusion

⚠️ **Real-world validation complexity**



Scenario library + simulation + progressive field validation

Strategies for overcoming these challenges

PROTOTYPE → PILOT → PRODUCTION

PROTOTYPE

Camera-based proof of concept: perception, short-term motion understanding and adaptive collision-free path planning.

PILOT

Multi-scenario validation on diverse Indian roads, with additional sensors and closed-loop simulation.

PRODUCTION

Vehicle-grade multi-sensor perception, real-time planning and simulation-backed validation under automotive safety constraints.

Potential impact on the target audience



Road users

Pedestrians, two-wheeler riders and passengers in mixed traffic



OEMs & ADAS teams

An Indian-road planning layer for driver assistance and autonomy



Fleets & public transport

Buses, taxis, autos and logistics on unstructured routes



Research & testing bodies

Scenario-based validation of adaptive navigation

Benefits of the solution (social, economic, environmental, etc.)

CURRENT CHALLENGE

Unstructured roads

+

Mixed traffic

+

Unpredictable movement

+

Limited lane information

India, 2022: 4,61,312 road accidents and 1,68,491 deaths (MoRTH)



PATHSENSE

Scene understanding

+

Motion prediction

+

Risk-aware path planning

+

Adaptive re-planning

Proposed system — benefits are expected outcomes, not measured results



EXPECTED BENEFITS



Safety

Better handling of unpredictable road interactions

Social



Mobility

Smoother, adaptive navigation where lanes are unreliable

Economic · Environmental



Scalability

Architecture can evolve toward multi-sensor vehicle systems

Economic



Indian-road relevance

Designed around mixed, irregular traffic behaviour

Social



Research value

A path to scenario-based validation of adaptive navigation

Research

Details / Links of the reference and research work

 Indian-road perception

- Varma et al., IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments, WACV 2019

 Path planning

- Paden et al., A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles, IEEE T-IV 2016
- González et al., A Review of Motion Planning Techniques for Automated Vehicles, IEEE T-ITS 2016

 Detection, tracking & depth

- Redmon et al., You Only Look Once, CVPR 2016
- Zhang et al., ByteTrack, ECCV 2022
- Yang et al., Depth Anything V2, NeurIPS 2024

 Motion prediction

- Mozaffari et al., Deep Learning-Based Vehicle Behavior Prediction for Autonomous Driving Applications: A Review, IEEE T-ITS 2022
- Alahi et al., Social LSTM, CVPR 2016

 Collision avoidance

- Fiorini & Shiller, Motion Planning in Dynamic Environments Using Velocity Obstacles, IJRR 1998
- Hayward, Near-miss determination through use of a scale of danger (time-to-collision), HRR 1972

 Autonomous-driving safety

- ISO 26262 — Road vehicles, functional safety
- ISO 21448 — Safety of the intended functionality (SOTIF)
- Shalev-Shwartz et al., On a Formal Model of Safe and Scalable Self-driving Cars, 2017

 Road context

- OpenStreetMap contributors — road network and points of interest
- MoRTH, Road Accidents in India 2022

 Problem statement & tools

- SIH 2026 problem statement SIH26037 — Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- MathWorks RoadRunner & Automated Driving Toolbox documentation